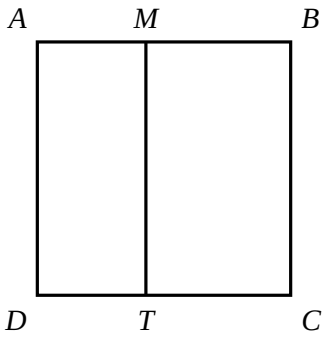
Preview: set-03/by_level/Grade_03-04/y2002_q06 (v1)

Perimeter of rectangles cut from a square or rectangle

Part A: Each question is worth 3 points

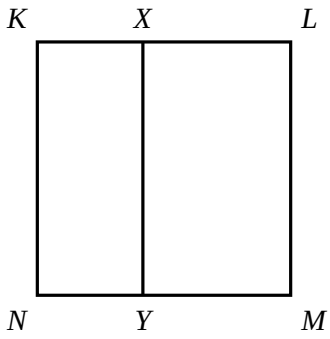
1. *ABCD* is a square. Its side is 7 cm. *AMTD* is a rectangle. Its shorter side is 3 cm.



How many centimetres is the perimeter of the square *ABCD* larger than that of the rectangle *AMTD*?

A. 14 cm ***B.*** 8 cm ***C.*** 7 cm ***D.*** 6 cm ***E.*** 4 cm

2. *KLMN* is a square. Its side is 12 cm. *KXYN* is a rectangle. Its shorter side is 5 cm.

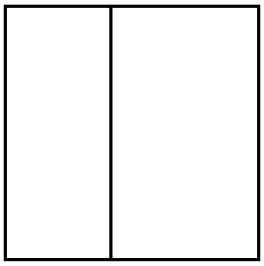


How many centimetres is the perimeter of the square *KLMN* larger than that of the rectangle *KXYN*?

A. 14 cm ***B.*** 24 cm ***C.*** 28 cm ***D.*** 31 cm ***E.*** 34 cm

Part B: Each question is worth 4 points

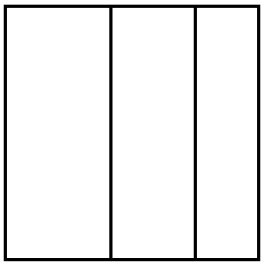
3. A square garden is surrounded by a fence 40 m long. The gardener divides it into 2 rectangular beds as shown and puts a separate fence around each bed.



How many metres of fence are around the beds altogether?

A. 60 m ***B.*** 56 m ***C.*** 50 m ***D.*** 47 m ***E.*** 40 m

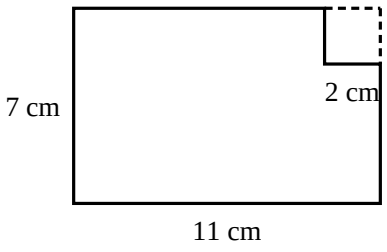
4. A square has perimeter 44 cm. It is cut along straight lines into 3 rectangles as shown.



What is the sum of the perimeters of all 3 rectangles?

A. 44 cm ***B.*** 66 cm ***C.*** 67 cm ***D.*** 77 cm ***E.*** 88 cm

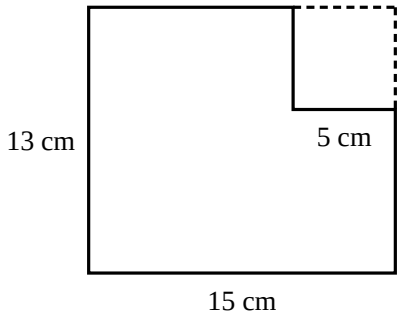
5. A rectangle is 11 cm long and 7 cm wide. A square with side 2 cm is cut off one corner.



What is the perimeter of the remaining shape?

A. 44 cm ***B.*** 40 cm ***C.*** 36 cm ***D.*** 32 cm ***E.*** 28 cm

6. A rectangle is 15 cm long and 13 cm wide. A square with side 5 cm is cut off one corner.

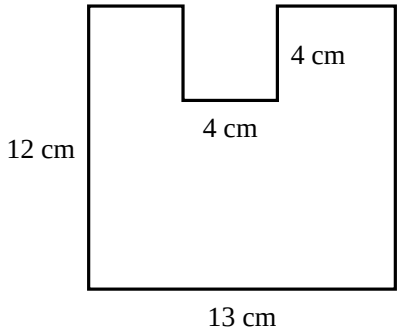


What is the perimeter of the remaining shape?

A. 36 cm ***B.*** 46 cm ***C.*** 51 cm ***D.*** 56 cm ***E.*** 66 cm

Part C: Each question is worth 5 points

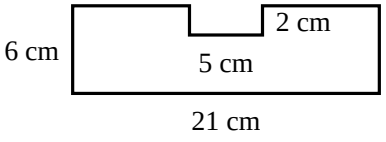
7. A 13 cm by 12 cm rectangle of cardboard has a rectangular notch 4 cm wide and 4 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 72 cm ***B.*** 67 cm ***C.*** 66 cm ***D.*** 58 cm ***E.*** 54 cm

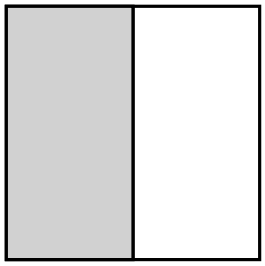
8. A 21 cm by 6 cm rectangle of cardboard has a rectangular notch 5 cm wide and 2 cm deep cut out of the middle of its top side.



What is the perimeter of the new shape?

A. 49 cm ***B.*** 58 cm ***C.*** 64 cm ***D.*** 67 cm ***E.*** 68 cm

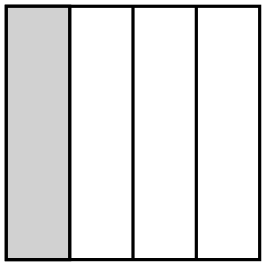
9. Mia cuts a square photo into 2 equal strips as shown. The perimeter of each strip is 36 cm.



How long is the side of the square?

A. 6 cm ***B.*** 9 cm ***C.*** 10 cm ***D.*** 12 cm ***E.*** 48 cm

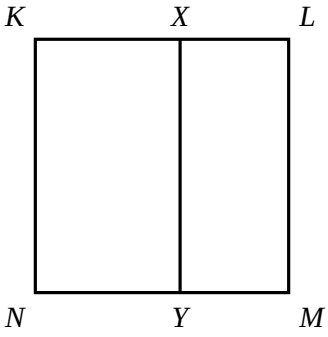
10. Mia cuts a square photo into 4 equal strips as shown. The perimeter of each strip is 50 cm.



What is the perimeter of the square?

A. 20 cm ***B.*** 60 cm ***C.*** 80 cm ***D.*** 100 cm ***E.*** 200 cm

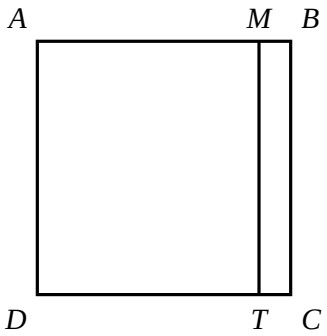
11. The square *KLMN* is cut into two rectangles. The perimeter of *KXYN* is 22 cm and the perimeter of *XLMY* is 20 cm.



What is the perimeter of the square *KLMN*?

A. 2 cm ***B.*** 28 cm ***C.*** 34 cm ***D.*** 35 cm ***E.*** 42 cm

12. The square *ABCD* is cut into two rectangles. The perimeter of *AMTD* is 30 cm and the perimeter of *MBCT* is 18 cm.



How long is the side of the square *ABCD*?

A. 5 cm ***B.*** 6 cm ***C.*** 7 cm ***D.*** 8 cm ***E.*** 24 cm

Answer Key

1. **B**	2. **A**	3. **A**	4. **E**	5. **C**	6. **D**
7. **D**	8. **B**	9. **D**	10. **C**	11. **B**	12. **D**

Solutions

1. B (8 cm) — Square: 4 × 7 = 28 cm. Rectangle: 3 + 7 + 3 + 7 = 20 cm. Difference 28 – 20 = 8 cm. [set-03/by_level/Grade_03-04/2002 Q6 · square_minus_strip]

2. A (14 cm) — Square: 4 × 12 = 48 cm. Rectangle: 5 + 12 + 5 + 12 = 34 cm. Difference 48 – 34 = 14 cm. [set-03/by_level/Grade_03-04/2002 Q6 · square_minus_strip]

3. A (60 m) — Side = 40 ÷ 4 = 10 m. There is 1 cut; each cut is 10 m long and is counted twice (once for each piece next to it): 40 + 1 × 2 × 10 = 60 m. The widths of the pieces don't matter! [set-03/by_level/Grade_03-04/2002 Q6 · pieces_total]

4. E (88 cm) — Side = 44 ÷ 4 = 11 cm. There are 2 cuts; each cut is 11 cm long and is counted twice (once for each piece next to it): 44 + 2 × 2 × 11 = 88 cm. The widths of the pieces don't matter! [set-03/by_level/Grade_03-04/2002 Q6 · pieces_total]

5. C (36 cm) — The two new edges (2 cm and 2 cm) replace the two removed pieces of the outer edge, which were also 2 cm and 2 cm. So the perimeter is the same as the rectangle's: 2 × (11 + 7) = 36 cm. [set-03/by_level/Grade_03-04/2002 Q6 · corner_cut]

6. D (56 cm) — The two new edges (5 cm and 5 cm) replace the two removed pieces of the outer edge, which were also 5 cm and 5 cm. So the perimeter is the same as the rectangle's: 2 × (15 + 13) = 56 cm. [set-03/by_level/Grade_03-04/2002 Q6 · corner_cut]

7. D (58 cm) — The bottom of the notch (4 cm) replaces the 4 cm taken out of the top edge, and the two sides of the notch add 2 × 4 cm. Perimeter = 2 × (13 + 12) + 2 × 4 = 58 cm. [set-03/by_level/Grade_03-04/2002 Q6 · notch]

8. B (58 cm) — The bottom of the notch (5 cm) replaces the 5 cm taken out of the top edge, and the two sides of the notch add 2 × 2 cm. Perimeter = 2 × (21 + 6) + 2 × 2 = 58 cm. [set-03/by_level/Grade_03-04/2002 Q6 · notch]

9. D (12 cm) — A strip is the full side long and 1/2 of it wide, so its perimeter is side + side + 2 × (side ÷ 2) = 6 × (side ÷ 2). 36 ÷ 6 = 6, so the side is 2 × 6 = 12 cm. [set-03/by_level/Grade_03-04/2002 Q6 · strips_to_square]

10. C (80 cm) — A strip is the full side long and 1/4 of it wide, so its perimeter is side + side + 2 × (side ÷ 4) = 10 × (side ÷ 4). 50 ÷ 10 = 5, so the side is 4 × 5 = 20 cm and the perimeter is 80 cm. [set-03/by_level/Grade_03-04/2002 Q6 · strips_to_square]

11. B (28 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 22 + 20 = 42 cm = 6 sides. Side = 42 ÷ 6 = 7 cm, perimeter 4 × 7 = 28 cm. [set-03/by_level/Grade_03-04/2002 Q6 · two_perimeters]

12. D (8 cm) — Together the two perimeters go around the square (4 sides) and along the cut twice (2 more sides): 30 + 18 = 48 cm = 6 sides. Side = 48 ÷ 6 = 8 cm. [set-03/by_level/Grade_03-04/2002 Q6 · two_perimeters]